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Linear and non-linear roll damping of a FPSO via system identification of a third order equation with sway-roll coupled damping effects

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ABSTRACT

This work focuses on roll decay experiments performed on hulls normally used as FPSOs. Considering the practical case of beam seas, Newton's second law and the Newton-Euler theorem lead to coupled sway, heave and roll equations for an arbitrary pole. Assuming symmetry, the heave equation is uncoupled, leaving the sway and roll equations coupled. The research shows that, even if inertia is decoupled, damping coupling is unavoidable and cannot be neglected. Nevertheless, a third order equation dealing only with roll can be constructed. The latter is used to develop a robust approach that uses system identification techniques to obtain damping properties from the decay tests. In this approach, non-linear damping is assessed using a cycle-by-cycle linear approach that uses system identification to match all the information obtained from the decay test. Both held-over and pre-oscillation tests are discussed, and new concept is presented: the coupled damping number (CDN), which relates the product of the main diagonal damping terms with coupling damping coefficients. The CDN is proven to be null for potential flows, and turns out to be very small even when viscous effects are present such as in decay tests, at least for typical FPSO hulls.

1. Introduction

Since the start of the use of FPSOs (Floating Production Storage and Offloading Platforms) in the 1960's, roll motions have revealed to be a major concern for operators of these units. Roll motion prediction for FPSOs are traditionally based on the knowledge established for conventional ships, and very few works oriented to issues specific to FPSOs have been published. Nonetheless, very large roll motions, with values above design predictions, have been reported for FPSOs over the last years, implying uncomfortable conditions for the crew, process plant downtime and degradation of the structural integrity of topside supports, hull, bilge keels and risers. It should be stressed that even though many similarities can be found between an FPSO and a ship, these structures differ in many aspects (lack of maneuvering capability; weight distribution with regard to topsides; and unconventional anti-roll system), which affects their roll behavior and places a question mark on the adequacy of currently employed methods for roll motion prediction.

In fact, numerous works related to the estimation of roll damping in oil tankers have been published during the last decades. Nevertheless, most have two fundamental issues which may propagate errors to damping estimation. The first is related to the lack of a unique definition of the pole necessary for rotational equations, despite the fact that the importance of this definition for damping estimation has been shown by some researchers. This pole is in principle arbitrary, as known from theory, and is known as either the “roll center” or the “center of roll”. However, these names prove to be meaningless, as they have no clear definition. A better understanding of the pole would definitely improve roll estimation procedures. The second fundamental issue is related to the understanding of the coupling of roll with other degrees of freedom. These two issues are addressed below.

Froude (1861) was one of the first to recognize the importance of roll motions for the operability of ships. Froude formulated roll damping in a linear plus quadratic velocity-dependent form to account for dissipation of energy during roll motion. He studied the effects of wave height and steepness on the rolling of ships and the influence of this phenomenon on the design of ship hull shape. Based on this work (Froude, 1861), he suggested designing the hull in such a way as to move the ship's natural roll frequency away from synchronization with the excitation waves. His work also supported the use of bilge keels to stabilize roll motion. Froude's method uses decay tests equating the potential energy lost to damping in each half cycle to the work done by the equivalent linearized damping moment during the same period. This produces an expression for the slope of the roll decay curve as a function of the linear and nonlinear damping coefficients. Later on, Dalzell (1976) extended Froude's method to the case of a linear-plus-

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cubic damping model.

Vugts (1968) conducted several experimental assessments to determine hydrodynamic coefficients of swaying, heaving and rolling cylinders in free fluid surfaces. His work has been used as the basis for numerous studies assessing the roll damping of floating bodies. Ikeda et al. (1978a) proposed a new technique to estimate roll damping based on decay tests. This technique separated roll damping into components, but ignored their interactions, and therefore the coupling of roll damping to other degrees of freedom was not taken into account (Ikeda et al., 1978a). Himeno (1981) provided a good review of roll damping phenomena in a very thorough literature survey of Japanese and worldwide literature on the subject. This review is largely based on the work of Ikeda et al. (1978b).

Faltinsen (1993) described another approach to predict damping. In his approach, energy loss is evaluated in terms of a linear equivalent damping coefficient, equating the energy loss of an equivalent sinusoidal behavior to the energy dissipation from a quadratic equation.

The vast majority of roll damping studies since Froude has focused on studying roll in a single degree of freedom, neglecting any coupling. But Ikeda et al. (1981) also conducted a number of forced sway and roll experiments. His paper deals with viscous effects on forces and moments acting on a ship in sway and roll motion at zero forward speed, and investigates roll damping and coupling terms in velocity-phase of roll into sway and sway into roll. From this work it can be concluded that damping coupling terms are large, having the same order of magnitude as the main diagonal dampings.

Bass and Haddara (1989, 1991) performed an extensive series of experiments to determine roll damping characteristics for a number of small fishing vessels. In the experiments, the models were attached to a dynamometer with only 2 degrees of freedom. The model was free to roll and heave, but restrained in all other modes. His investigations included not only roll damping but also roll-sway coupled damping. He clearly showed the influence of roll-sway and sway-roll coupling on roll damping, which has a non-linear dependence on the position of the roll axis (the axis on which the model is free to rotate).

Standing (1991) also conducted many model tests to investigate viscous roll damping and the effect of sway and heave motions on the roll response of a transportation barge. Numerical examples suggest that the roll and sway equations may be decoupled if the motions are defined relative to a certain 'roll center'. His work assumes that the roll, sway and heave equations are coupled only through the relative velocity term in the roll damping moment. The equations are otherwise decoupled by defining them relative to the 'roll center'. This 'roll center', however, is not clearly defined.

During the past 40 years much more information has become available supporting that roll damping prediction is influenced by the position of the 'roll center'. The first serious discussions emerged during the 1980s, when Ikeda, 2004 carried out forced roll tests on four fishing boat models. He observed that lowering the position of the rotation axis of the forced test (the 'roll center') led to decreased roll damping. Bass and Haddara (1989) showed that the equivalent non-dimensional linear roll damping coefficient increases along with the distance between the position of the rotation axis ('roll center') and the center of gravity. Chun et al. (2001) arrived at the same conclusion. Park et al. (2000) studied the effects of the position of the rotation axis (again called the 'roll center') on damping for FPSO sections using decay experiments. They also considered a symmetric section of an FPSO and positioned the rotation axis on the symmetry plane. Their findings indicate that roll damping is sensitive to the position of the axis ('roll center') on the symmetry plane. Roll damping will increase if the rotation axis is placed above the still-water level.

Thus far, previous studies have confirmed the importance of the position of the rotation axis. However, for a freely floating body there is no fixed rotation axis, hence a clear definition of the 'roll center' is not traceable in the literature, and in effect each researcher has his/her own interpretation. In the classical Principles of Naval Architecture - PNA a quote states that "a very rough estimate for the location of the roll center is halfway between the center of gravity and the center of buoyancy", implying that the 'roll center' lies on the symmetry plane.

Overall, all relevant studies of roll motion highlight the need for a better understanding of the 'roll center'. Different definitions may be found: 'center of gravity' (Vugts, 1968), 'still-water level', 'center of buoyancy' and others have all been considered as the 'roll center'. This difficulty can even be found in Ikeda's formulations: in Ikeda et al. (1978a) and Ikeda et al. (1977) the quantity known as OG represents the distance from the still-water level to the roll axis (vertical position of the 'roll center'). In other papers, such as Ikeda (1984) and Ikeda et al. (1993), OG is defined as the distance from the still-water level to the center of gravity.

Stewart et al. (1979) mentioned an instantaneous 'roll center'. According to his work, the roll center is instantaneously at rest as the vessel moves through waves, but tends to wander wildly, and may move from well above to well below the vessel in irregular waves. The main reason why Stewart's formulation was not taken up by other researchers is not very clear. Roberts and Dacunha (1985) identified the 'roll center' as a point where the roll equation decouples from sway, which is in fact impossible since damping coupling still remains (see below). Later on, Standing (1991) proposed the following definition: "When there is significant damping, the 'roll center' may be defined as the point of minimum sway response in a forced-roll or free-decay type of experiment or simulation, when a pure moment is applied". This is an obscure sentence, at least for the present authors. In both references, the position of the 'roll center' is assumed to be constant and independent of frequency. The mismatch of 'roll center' definitions may have motivated Molin (2002) to call it the "mystérieux center du roulis" (mysterious roll center, in French). Recently, Fernandes et al. (2015, 2016) have noticed that the instantaneous center of rotation (ICR), that is, the point with zero velocity, varies in time, as predicted by Newtonian mechanics. Fernandes et al. (2015, 2016) also proposed the concept of a "Most Often Instantaneous Roll Center" – MOIRC value, as also discussed below. Of course this MOIRC value could be called the 'roll center', but the authors see no reason to do that. In fact, the 'roll center' (and the 'center of roll', for that matter) seems indeed to be a fantasy that does not help those working with roll damping estimation.

Based on the challenges described above, this work presents an innovative, robust approach to assess roll damping of FPSOs, taking into account a correct definition of the pole on which to center the motion equations as well as the influence of the unavoidable coupled damping for sway and roll damping estimation, using both numerical and experimental techniques.

2. Third order equation

The motion of a floating body exposed to external forces such as wind, wave and currents takes place in six degrees of freedom. The motion equations can be derived using the Newton-Euler or Lagrange equations. Clayton and Bishop (1982) expressed general, fully non-linear equations of rigid-body kinetics in a vectorial framework. In order to construct the governing motion equations, the following assumptions and simplifications are made:

(i) The floating body is slender and rigid, with a symmetry plane;
(ii) Motion amplitude is small, so that equations can be linearized;
(iii) The effects of viscosity are neglected except for roll motion;

A right handed co-ordinate system (x, y, z) is considered with respect to an arbitrary position O, as shown in Fig. 1. The translation displacements in x, y and z directions with respect to the origin, denoted by η_1, η_2 and η_3 indicate surge, sway and heave, respectively. The angular displacements about the same set of axis, η_4, η_5 and η_6, indicate roll, pitch and yaw, respectively.

Under the above assumptions, three linearly coupled differential

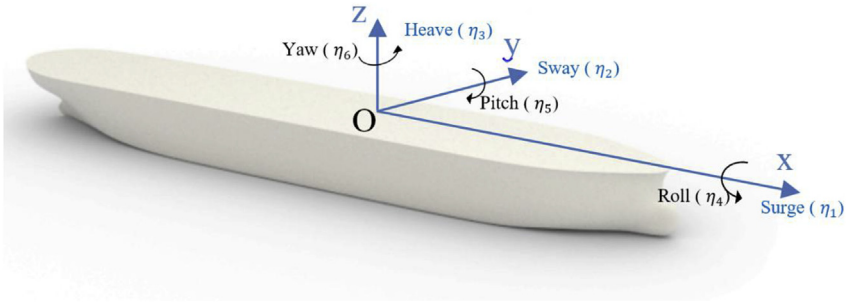


Fig. 1. Coordinate system with an arbitrary origin 'O'.

homogeneous equations of motion in a vertical plane for a free roll decay test (imposing an initial roll displacement on a freely floating body) can be written as:

$$\sum_{j=2}^{4}\sum_{k=2}^{4}\{([M_{jk}]+[A_{jk}])\ddot{\eta}_k+[B_{jk}]\dot{\eta}_k+[C_{jk}]\eta_k\}=0 \tag{2.1}$$

where $[M_{jk}]$ are the components of the generalized mass matrix of the ship and z_c is the vertical distance of the pole to the center of gravity, $[A_{jk}]$ and $[B_{jk}]$ are the frequency-dependent added-mass and damping coefficients, respectively and $[C_{jk}]$ are the hydrostatic restoring coefficients. Due to symmetry, the generalized forms of mass, added-mass and damping matrix coefficients are given by:

$$[M_{jk}]=\begin{pmatrix} M & 0 & -Mz_c \\ 0 & M & 0 \\ -Mz_c & 0 & I_{44}\end{pmatrix}[A_{jk}]=\begin{pmatrix} A_{22} & 0 & A_{24} \\ 0 & A_{33} & 0 \\ A_{24} & 0 & A_{44}\end{pmatrix}[B_{jk}]$$
$$=\begin{pmatrix} B_{22} & 0 & B_{24} \\ 0 & B_{33} & 0 \\ B_{24} & 0 & B_{44}\end{pmatrix}[C_{jk}]=\begin{pmatrix} 0 & 0 & 0 \\ 0 & C_{33} & 0 \\ 0 & 0 & C_{44}\end{pmatrix}$$

It is well known that the heave equation entirely decouples due to symmetry, with the sway-roll Equations remaining coupled (2.2) (Vugts, 1968).

$$\begin{pmatrix} M+A_{22} & A_{24}-Mz_c \\ A_{42}-Mz_c & I_4+A_{44}\end{pmatrix}\begin{pmatrix}\ddot{\eta}_2 \\ \ddot{\eta}_4\end{pmatrix}+\begin{pmatrix} B_{22} & B_{24} \\ B_{42} & B_{44}\end{pmatrix}\begin{pmatrix}\dot{\eta}_2 \\ \dot{\eta}_4\end{pmatrix}+\begin{pmatrix} 0 & 0 \\ 0 & C_{44}\end{pmatrix}\begin{pmatrix}\eta_2 \\ \eta_4\end{pmatrix}=\begin{pmatrix}0 \\ 0\end{pmatrix} \tag{2.2}$$

It is easy to inertially decouple the roll and sway equations by moving the origin of the coordinate system to a position at which the mass matrix is diagonal (zero off-diagonal elements). This position of the origin is referred to here as the Inertial Decoupling Point (IDP). A new set of sway and roll equations can then be written for the IDP coordinate system as (2.3):

$$\begin{pmatrix} M+A_{22} & 0 \\ 0 & I_4^{idp}+A_{44}^{idp}\end{pmatrix}\begin{pmatrix}\ddot{\eta}_2 \\ \ddot{\eta}_4\end{pmatrix}+\begin{pmatrix} B_{22} & B_{24}^{idp} \\ B_{42}^{idp} & B_{44}^{idp}\end{pmatrix}\begin{pmatrix}\dot{\eta}_2 \\ \dot{\eta}_4\end{pmatrix}+\begin{pmatrix} 0 & 0 \\ 0 & C_{44}^{idp}\end{pmatrix}\begin{pmatrix}\eta_2 \\ \eta_4\end{pmatrix}=\begin{pmatrix}0 \\ 0\end{pmatrix} \tag{2.3}$$

The vertical position of the IDP (z_{idp}) which is related to the vertical position of the center of gravity (Zc.g.) and the coupling added mass ($A_{42}^{c.g.}$) can be easily obtained by the Formula (2.4).

$$z_{idp}=z_{c.g.}-\left(\frac{A_{42}^{c.g.}}{M+A_{22}}\right) \tag{2.4}$$

Note that, despite the roll being inertially decoupled, non-zero coupled damping terms, B_{24}^{idp} and B_{42}^{idp}, still remain. These terms are by no means small, as shown below. Indeed, these terms are numerically large, having the same order of magnitude as B_{22} and B_{44}^{idp}. So the quote from the PNA (Lewis, 1989) stating that "roll is still coupled to the sway equation, but only through the coefficient B_{42}, and this coupling is not strong (…) (therefore) we ignore the coupling terms" does not seem to be correct.

Another relevant point is related to Equation (2.5), which corresponds to the second line of Equation (2.3).

$$(I_4^{idp}+A_{44}^{idp})\ddot{\eta}_4+B_{42}^{idp}\dot{\eta}_2+B_{44}^{idp}\dot{\eta}_4+C_{44}^{idp}\eta_4=0 \tag{2.5}$$

This is the roll equation, and it is clear that it includes a term depending on sway velocity and a damping coupling term. However, for decades the equation for one degree of freedom has been used in the literature (recall the PNA quote mentioned above - Lewis, 1989), ignoring this coupling-dependent term (2.6)

$$\ddot{\eta}_4+B(\dot{\eta}_4)\dot{\eta}_4+K(\eta_4)\eta_4=0 \tag{2.6}$$

Note that Equation (2.6) has no theoretical basis unless $B_{42}^{idp}\dot{\eta}_2=0$ for some reason. In fact, during decay tests (especially for the pure moment method – see below) the sway velocity might be small, but the coupling damping term is not, as shown below. Even if this were the case, the pole of the Equation with one degree of freedom (defined according to Equation (2.4)) can only be the IDP - it definitely cannot be the CG (Center of Gravity) or another point. One can find in the literature studies that place the pole at the CG, then use the parallel axis theorem to get the damping with respect to the IDP, which is conceptually wrong.

Now, note that the sway-roll problem is semi-definite, since the sway mode has no restoring force. Hence, the system is initially of fourth order, as seen in Equation (2.2), but it may become third order if the pole is positioned on the IDP. In fact, as shown in Appendix A, the system of Equation (2.3) may be written as:

$$\{I_4^{idp}+A_{44}^{idp}\}\ddot{\eta}_4+\left\{\frac{B_{22}(I_4^{idp}+A_{44}^{idp})}{(M+A_{22})}+B_{44}^{idp}\right\}\ddot{\eta}_4$$
$$+\left\{\frac{B_{22}B_{44}^{idp}-B_{24}^{idp}B_{42}^{idp}}{(M+A_{22})}+C_{44}^{idp}\right\}\dot{\eta}_4+\left\{\frac{B_{22}C_{44}^{idp}}{(M+A_{22})}\right\}\eta_4=0 \tag{2.7}$$

$$\{I_4^{idp}+A_{44}^{idp}\}\ddot{\eta}_2+\left\{\frac{B_{22}(I_4^{idp}+A_{44}^{idp})}{(M+A_{22})}+B_{44}^{idp}\right\}\ddot{\eta}_2+\left\{\frac{B_{22}B_{44}^{idp}-B_{24}^{idp}B_{42}^{idp}}{(M+A_{22})}\right\}\dot{\eta}_2$$
$$=\left\{\frac{B_{22}C_{44}^{idp}}{(M+A_{22})}\right\}\eta_4 \tag{2.8}$$

Note that Equation (2.7) is homogeneous, involving only roll. The coupling terms add to the roll restoring term, which multiplies $\dot{\eta}_4$. Hence, the remainder of this work will deal with Equation (2.7) to address the roll problem on an innovative, robust basis. Also note that, in principle, once η_4 is know it can be used to obtain $\dot{\eta}_2$ and plug it into Equation (2.8), allowing η_2 to be obtained after integration.

A comprehensive 3rd order roll equation including the coupling terms is presented here. The 3rd order equation is used in a robust method to assess roll damping in the non-linear quadratic case, including the influence of coupling terms.

3. Linear system identification

Equation (2.7) can be written in dimensionless form as:

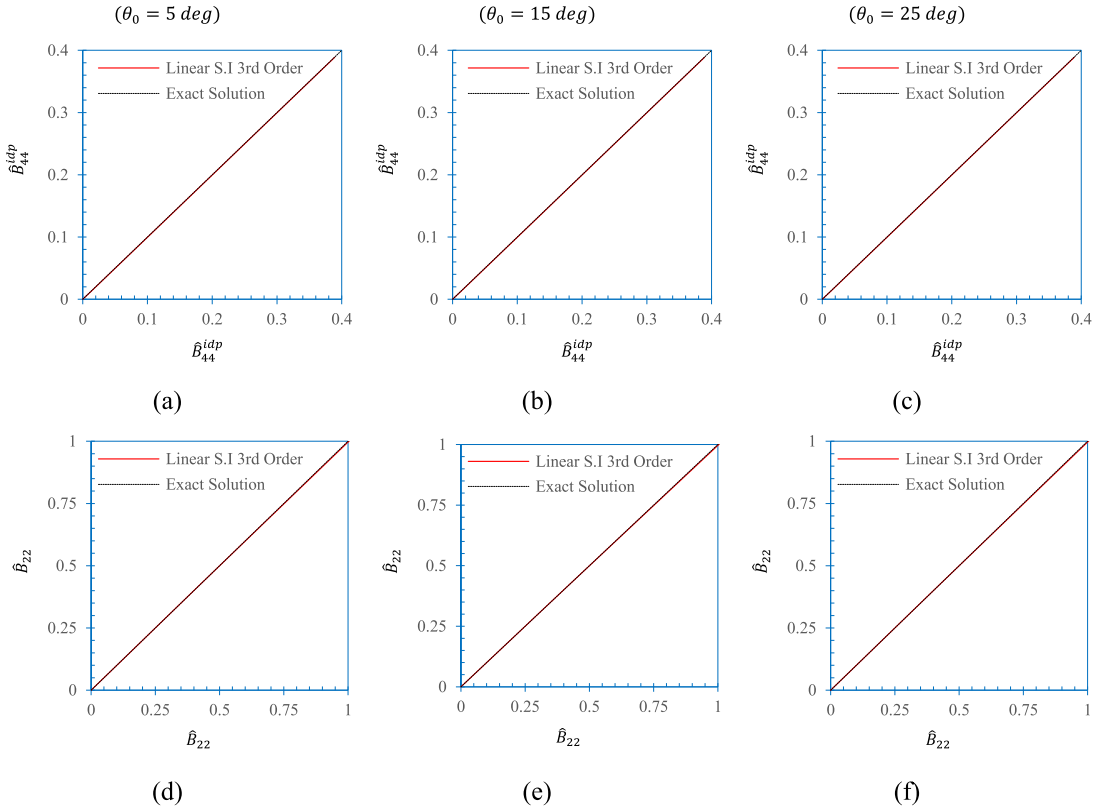


Fig. 2. Linear roll damping estimation by the System Identification Method (SI) for different initial angles.

$$\ddot{\eta}_4^* + \{\hat{B}_{22} + \hat{B}_{44}^{idp}\}\ddot{\eta}_4^* + \{\hat{B}_{22}\hat{B}_{44}^{idp} - \hat{B}_{24}^{idp}\hat{B}_{42}^{idp} + \hat{C}_{44}^{idp}\}\dot{\eta}_4^* + \{\hat{B}_{22}\hat{C}_{44}^{idp}\}\eta_4^* = 0, \tag{3.1}$$

where

$$\hat{B}_{22} = \frac{B_{22}}{(M + A_{22}).\,\omega_n}\,\hat{B}_{44}^{idp} = \frac{B_{44}^{idp}}{(I_4^{idp} + A_{44}^{idp}).\,\omega_n}\,\hat{C}_{44}^{idp}$$
$$= \frac{C_{44}^{idp}}{(I_4^{idp} + A_{44}^{idp}).\,\omega_n^2}\,\hat{B}_{24}^{idp} = \frac{B_{24}^{idp}}{(M + A_{22}).\,L.\,\omega_n}\,\hat{B}_{42}^{idp} = \frac{B_{42}^{idp}.\,L}{(I_4^{idp} + A_{44}^{idp}).\,\omega_n},$$

where L is a length unit, and the non-dimensional sway and roll and their derivatives can be expressed as below:

$$\eta_4^* = \eta_4 \quad \eta_2^* = \frac{\eta_2}{L}$$
$$\dot{\eta}_4^* = \frac{\dot{\eta}_4}{\omega_n} \quad \dot{\eta}_2^* = \frac{\dot{\eta}_2}{L.\omega_n}$$
$$\ddot{\eta}_4^* = \frac{\ddot{\eta}_4}{\omega_n^2} \quad \ddot{\eta}_2^* = \frac{\ddot{\eta}_2}{L.\omega_n^2}$$

Looking for a particular solution of the form $\eta_4^* = ke^{\lambda^* t^*}$, substituting η_4^* and its derivatives into Equation (3.1), one obtains

$$(\hat{B}_{22} + \hat{B}_{44}^{idp})\lambda_i^{*2} + (\hat{B}_{22}\hat{B}_{44}^{idp} - \hat{B}_{24}^{idp}\hat{B}_{42}^{idp})\lambda_i^* + \hat{B}_{22}\hat{C}_{44}^{idp} = -\lambda_i^*(\lambda_i^{*2} + \hat{C}_{44}^{idp}), \tag{3.2}$$

where $\lambda_i^* = \frac{\lambda_i}{\omega_n}$,. A particular solution of λ_i^* for $i = 1, 2$ and 3 can be obtained from Equation (3.2), taking into account the initial conditions k_1, k_2 and k_3. Considering free roll decay, the general solution for the linear roll response can be represented in exponential form, as seen below:

$$\eta_4^*(t^*) = k_1 e^{\lambda_1^* t^*} + k_2 e^{\lambda_2^* t^*} + k_3 e^{\lambda_3^* t^*} \tag{3.3}$$

The linear equivalent coefficients of the roll Equation (3.1) can be obtained by applying a linear system identification technique to every cycle of the roll time series. Linear system identification assumes that the roll response has a form such as that of Equation (3.3) for a cycle (peak-to-peak or trough-to-trough) with period T, and proceeds by applying non-linear least squares using the Levenberg-Marquardt method (Levenberg, 1944; Marquardt, 1963; Moré, 1978) to the above equation in order to obtain k_i and λ_i^*. The coefficients of the 3rd order equation (3.1) can be obtained by providing the correct values of k_i and λ_i^* obtained from linear system identification for the following initial conditions:

$$\eta_4^*(0) = \theta_0$$
$$\dot{\eta}_4^*(0) = 0$$
$$\ddot{\eta}_4^*(0) = -\hat{C}_{44}^{idp}\theta_0$$

With known k_i and λ_i^*, the following system of equations can be written based on equation (3.2):

$$\begin{pmatrix} 1 & \lambda_1^{*2} & \lambda_1^* \\ 1 & \lambda_2^{*2} & \lambda_2^* \\ 1 & \lambda_3^{*2} & \lambda_3^* \end{pmatrix} \begin{pmatrix} \hat{B}_{22}\hat{C}_{44}^{idp} \\ \hat{B}_{22} + \hat{B}_{44}^{idp} \\ \hat{B}_{22}\hat{B}_{44}^{idp} - \hat{B}_{24}^{idp}\hat{B}_{42}^{idp} \end{pmatrix} = \begin{pmatrix} -\lambda_1^*(\lambda_1^{*2} + \hat{C}_{44}^{idp}) \\ -\lambda_2^*(\lambda_2^{*2} + \hat{C}_{44}^{idp}) \\ -\lambda_3^*(\lambda_3^{*2} + \hat{C}_{44}^{idp}) \end{pmatrix} \tag{3.4}$$

$\hat{C}_{44}^{idp}$ is known from the initial conditions $\eta_4^*(0)$ and $\ddot{\eta}_4^*(0)$:

$$\hat{C}_{44}^{idp} = -\left(\frac{\lambda_1^{*2}k_1 + \lambda_2^{*2}k_2 + \lambda_3^{*2}k_3}{k_1 + k_2 + k_3}\right) \tag{3.5}$$

Therefore, $\hat{B}_{22}$, $\hat{B}_{44}^{idp}$ and $\hat{B}_{24}^{idp}\hat{B}_{42}^{idp}$ can be obtained from Equation (3.4). Note that this is a procedure that gets all the information available from the decay test, which is another proposal of the present work. This is unlike the Froude and Faltinsen methods, which use global properties: the potential energy difference in the former, and the equivalent harmonic energy dissipation in the latter.

4. Linear validation

To validate the SI (System Identification method), time series were generated based on the coefficients of numerical model for different initial angles (θ_0) and linear damping coefficients provided by the

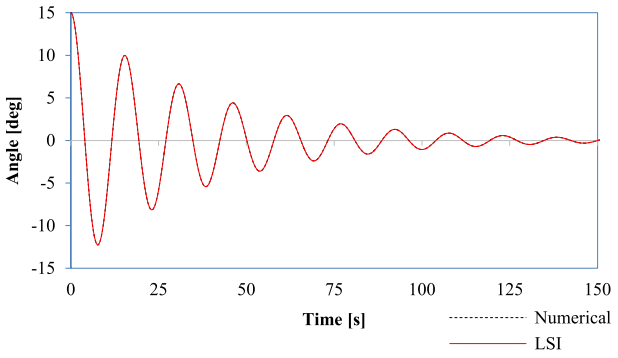


Fig. 3. Comparison of the roll time series from the numerical model and the results obtained from linear system identification (LSI).

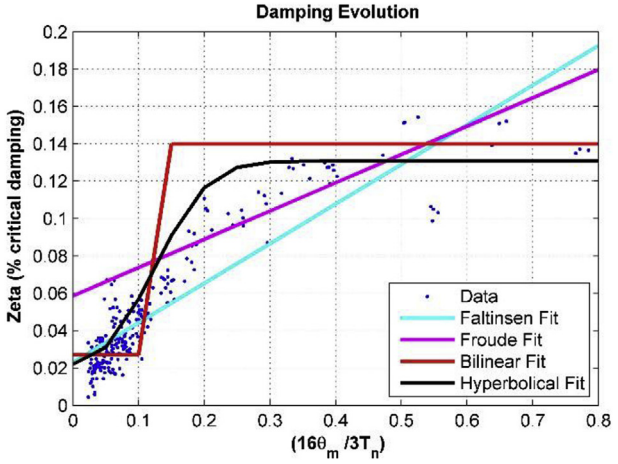


Fig. 4. Typical damping plot methods to obtain linear and quadratic terms of roll damping, where $\varsigma = \frac{\hat{B}_{44}}{\hat{B}_{44}^{critical}}$; (Oliveira and Fernandes, 2014); see reference for definitions.

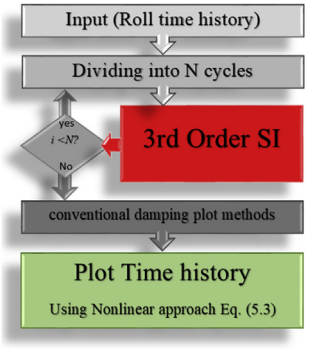


Fig. 5. System identification Flowchart.

general solution of equations with the IDP as the origin. Linear dampings are extracted from the time series by applying the SI and compared with exact numerical solutions in Fig. 2.

It has been observed that the non-dimensional order of magnitude of $\hat{B}_{24}^{idp}\hat{B}_{42}^{idp}$ is much smaller than those of $\hat{B}_{22}$ and $\hat{B}_{44}^{idp}$ and, consequently, SI estimates of this value suffer from unavoidable numerical errors. For instance, in the case studied here, $\hat{B}_{24}^{idp}\hat{B}_{42}^{idp}$ equals 1.8e-03, while the value estimated with SI is 1.6e-03. By identifying the role of $\hat{B}_{24}^{idp}\hat{B}_{42}^{idp}$ on the roll equation, it has been concluded that this numerical error (2e-04) may be negligible. Indeed, Fig. 3 illustrates that the roll time series obtained by linear system identification is very close to the direct numerical result.

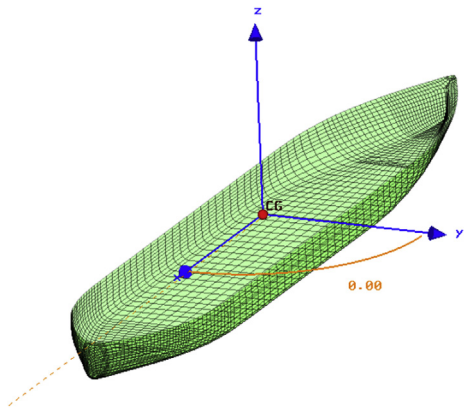


Fig. 6. Numerical model of the FPSO; draft = 14.70 m.

Table 1
Particulars of the numerical model.

Properties	
Displacement	211873.800 ton
Length Overall	337.359 m
Beam	54.50 m
Depth	27.80 m
Reference Draft	14.70 m
KB	7.56 m
KG	14.31 m
Transverse GM	9.05 m
Longitudinal GM	471.15 m
Rxx	20.52 m
Ryy	73.57 m
Rzz	74.05 m

Table 2
Numerical test conditions.

$\frac{\hat{B}_{44_q}^{idp}}{\hat{B}_{44_l}^{idp}}$:		θ_0:
	0.00	
	0.10	*5*
	0.25	*10*
	0.50	*15*
	1.00	*20*
	2.00	*25*
	3.00	*30*
	5.00	

Table 3
SIs with (1) and without (2) taking into consideration the term. $(\hat{B}_{24}^{idp}\hat{B}_{42}^{idp})$

1) 3rd SI $(\hat{B}_{24}^{idp}\hat{B}_{42}^{idp} \neq 0)$:

$$\begin{pmatrix} 1 & \lambda_1^{*2} & \lambda_1^* \\ 1 & \lambda_2^{*2} & \lambda_2^* \\ 1 & \lambda_3^{*2} & \lambda_3^* \end{pmatrix}\begin{pmatrix} \hat{B}_{22}\hat{C}_{44}^{idp} \\ \hat{B}_{22}+\hat{B}_{44}^{idp} \\ \hat{B}_{22}\hat{B}_{44}^{idp}-\hat{B}_{24}^{idp}\hat{B}_{42}^{idp} \end{pmatrix} = \begin{pmatrix} -\lambda_1^*(\lambda_1^{*2}+\hat{C}_{44}^{idp}) \\ -\lambda_2^*(\lambda_2^{*2}+\hat{C}_{44}^{idp}) \\ -\lambda_3^*(\lambda_3^{*2}+\hat{C}_{44}^{idp}) \end{pmatrix}$$

2) 3rd SI $(\hat{B}_{24}^{idp}\hat{B}_{42}^{idp} = 0)$:

$$\begin{pmatrix} 1 & \lambda_1^{*2} & \lambda_1^* \\ 1 & \lambda_2^{*2} & \lambda_2^* \\ 1 & \lambda_3^{*2} & \lambda_3^* \end{pmatrix}\begin{pmatrix} \hat{B}_{22}\hat{C}_{44}^{idp} \\ \hat{B}_{22}+\hat{B}_{44}^{idp} \\ \hat{B}_{22}\hat{B}_{44}^{idp} \end{pmatrix} = \begin{pmatrix} -\lambda_1^*(\lambda_1^{*2}+\hat{C}_{44}^{idp}) \\ -\lambda_2^*(\lambda_2^{*2}+\hat{C}_{44}^{idp}) \\ -\lambda_3^*(\lambda_3^{*2}+\hat{C}_{44}^{idp}) \end{pmatrix}$$

5. Non linear approach

By applying SI, the linear equivalent damping and restoring coefficients could be obtained for each cycle (with their corresponding mean angles). This study assumes that the nonlinearity of roll damping can be represented as a linear-plus-quadratic form, as below:

$$\hat{B}_{44}^{idp}(\dot{\eta}_4^*) = \hat{B}_{44_l}^{idp} + \hat{B}_{44_q}^{idp}|\dot{\eta}_4^*| \tag{5.1}$$

The linear $(\hat{B}_{44_l}^{idp})$ and quadratic $\left(\hat{B}_{44_q}^{idp}\right)$ parts of roll damping can be

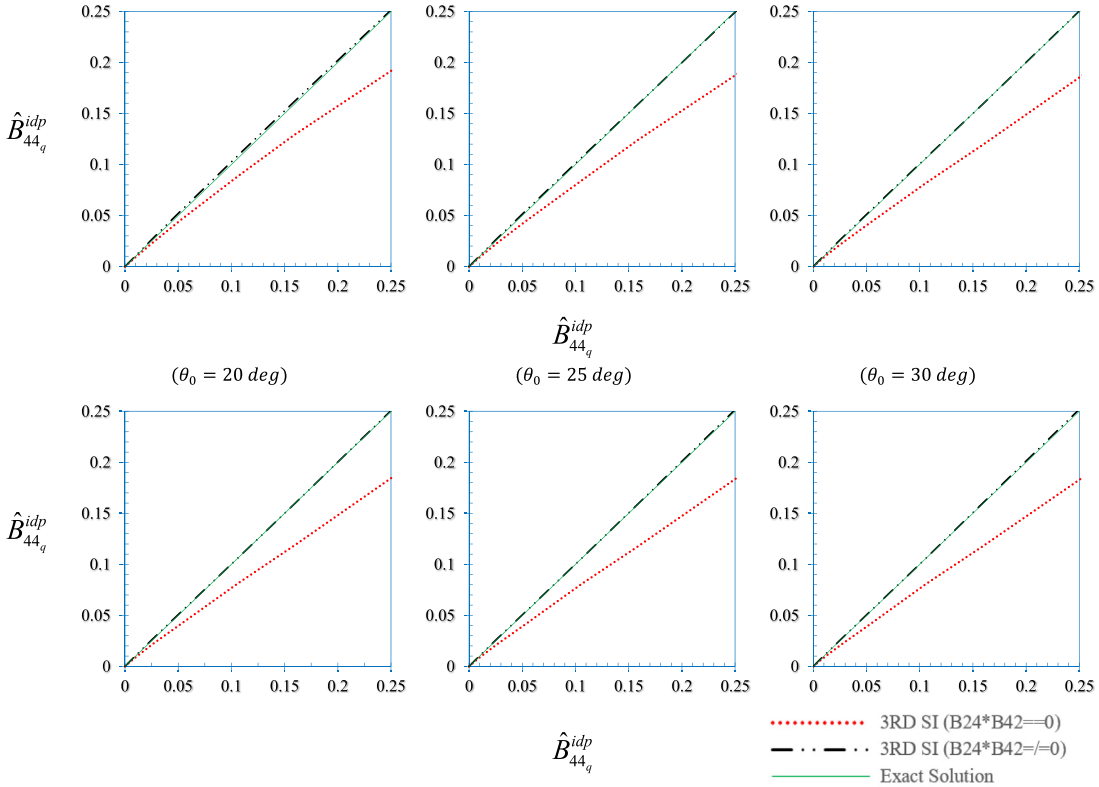


Fig. 7. Roll quadratic damping $\left(\hat{B}^{idp}_{44_q}\right)$ estimation by S.I with and without influence of $\hat{B}^{idp}_{24}$ and. $\hat{B}^{idp}_{42}$.

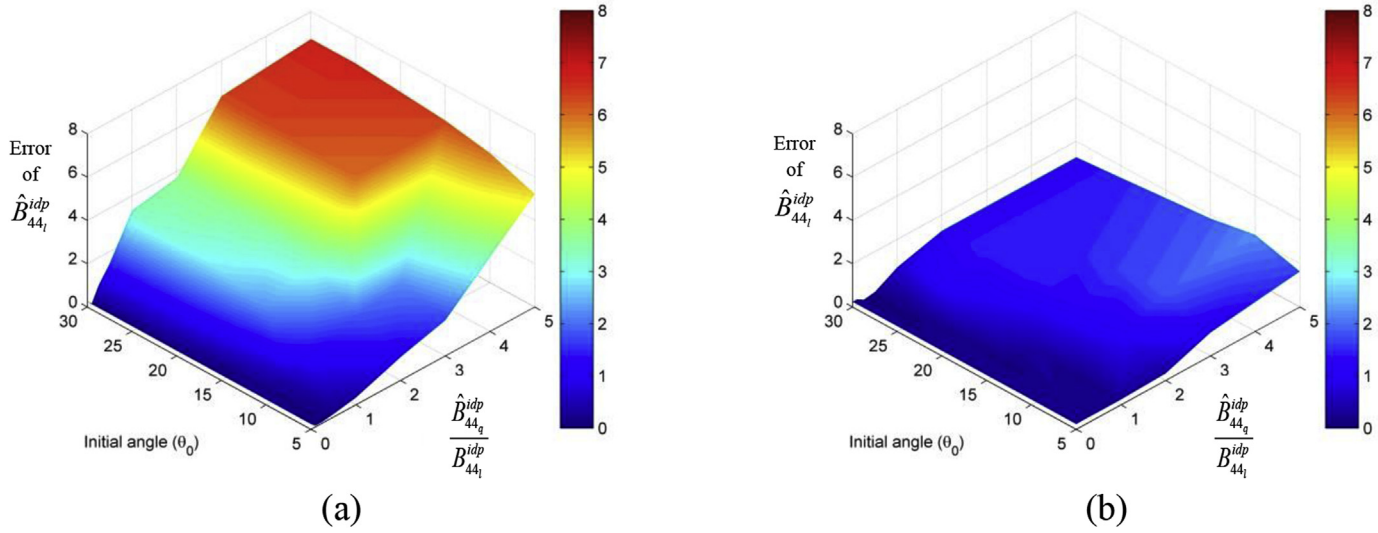


Fig. 8. S.I. errors for the linear term of roll damping: (a) 3rd SI ($\hat{B}^{idp}_{24}\hat{B}^{idp}_{42} = 0$) (b) 3rd SI. ($\hat{B}^{idp}_{24}\hat{B}^{idp}_{42} \neq 0$).

extracted from conventional damping plot methods such as those described by Froude, Faltinsen and others. See for example Fig. 4, extracted from Oliveira and Fernandes (2014).

Considering the derivative of $\hat{B}^{idp}_{44}(\dot{\eta}^*_4)$.

$$\frac{d}{dt}\{\hat{B}^{idp}_{44}(\dot{\eta}^*_4).\ \dot{\eta}^*_4\} = \hat{B}^{idp}_{44_l}\ddot{\eta}^*_4 + 2 \times \hat{B}^{idp}_{44_q}|\dot{\eta}^*_4|\dot{\eta}^*_4 \tag{5.2}$$

The 3rd order roll equation including linear-plus-quadratic damping terms can be obtained from Equation (5.3) as:

$$\ddot{\eta}^*_4 + \{\hat{B}_{22} + \hat{B}^{idp}_{44_l}\}\ddot{\eta}^*_4 + 2 \times \hat{B}^{idp}_{44_q}|\dot{\eta}^*_4|\ddot{\eta}^*_4 + \hat{B}_{22}\hat{B}^{idp}_{44_q}|\dot{\eta}^*_4|\dot{\eta}^*_4 + \{\hat{B}_{22}\hat{B}^{idp}_{44} - \hat{B}^{idp}_{24}\hat{B}^{idp}_{42}\}\dot{\eta}^*_4 + \hat{C}^{idp}_{44}\dot{\eta}^*_4 + \hat{B}_{22}\hat{C}^{idp}_{44}\eta^*_4 = 0 \tag{5.3}$$

Substituting the obtained coefficients from linear system identification into equation (5.3), the roll time series can be obtained by solving the equation using the Fourth Order Runge-Kutta method. The system identification algorithm can be described by the illustrated flowchart in Fig. 5.

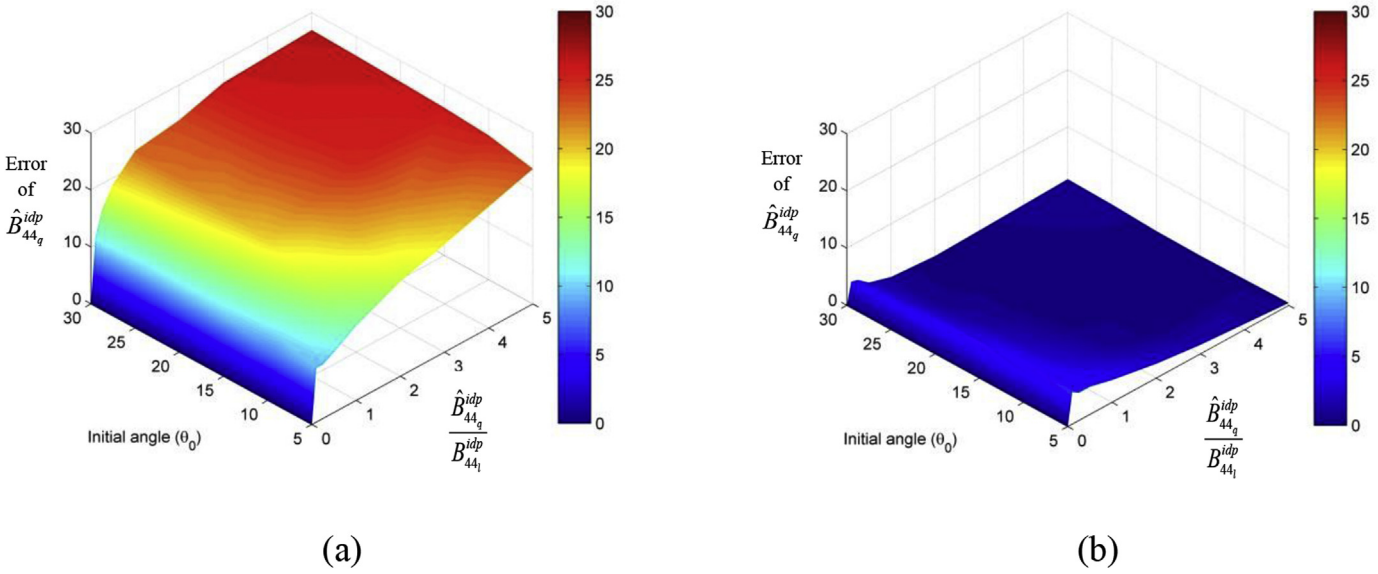


Fig. 9. S.I. errors for the quadratic term of roll damping:(a) 3rd LSI ($\hat{B}_{24}^{idp}\hat{B}_{42}^{idp} = 0$) (b) 3rd LSI. ($\hat{B}_{24}^{idp}\hat{B}_{42}^{idp} \neq 0$).

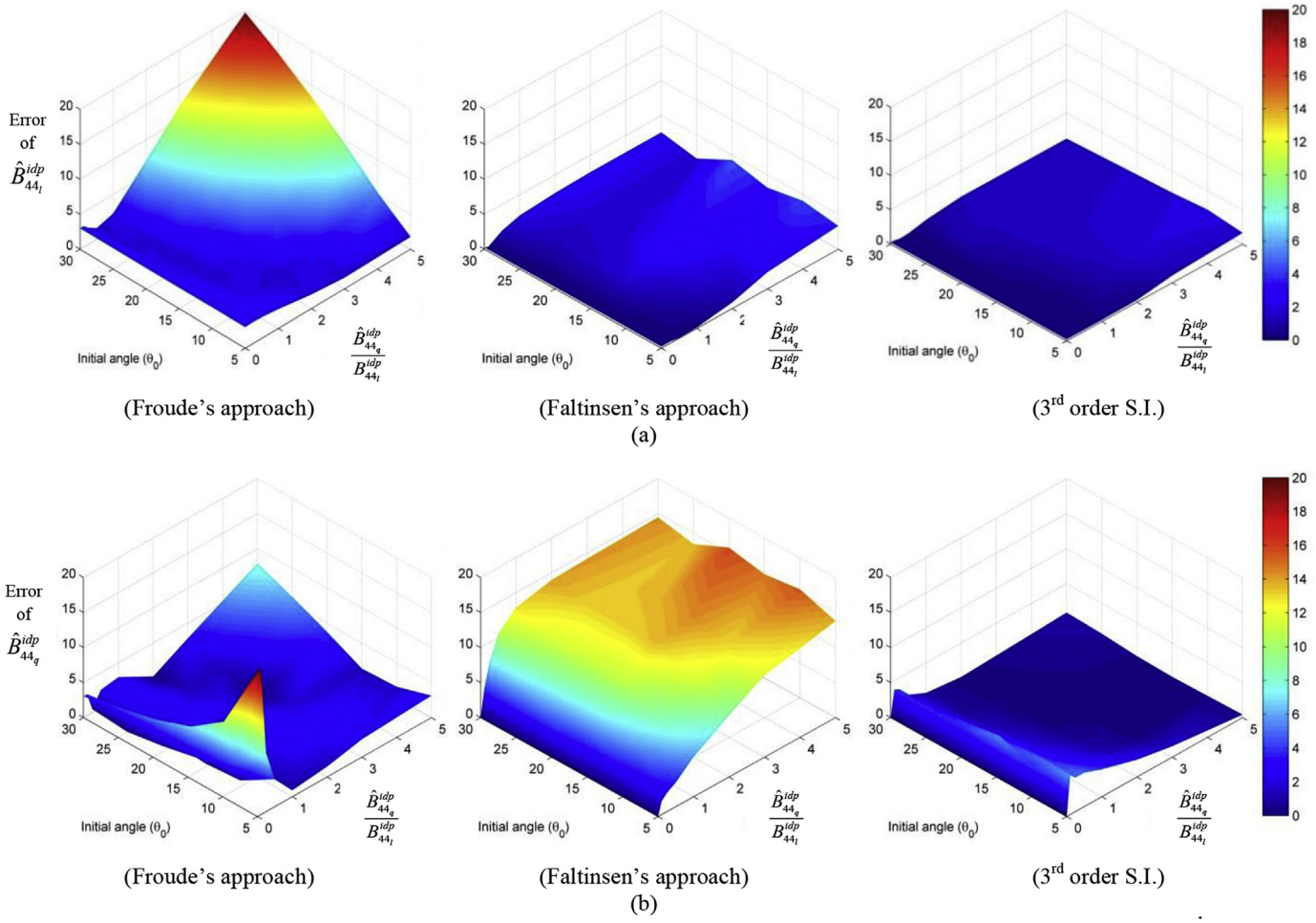


Fig. 10. Comparison of the errors of Froude's and Faltinsen's approaches and the errors of a 3rd order SI for:(a) Linear damping term $\hat{B}_{44_l}^{idp}$ (b) Quadratic damping term. $\hat{B}_{44_q}^{idp}$.

Table 4
Comparison between approaches.

	3rd S.I.	Faltinsen	Froude
Initial Angle sensitivity	*	**	***
Nonlinear Ratio sensitivity	*	**	***
Precision of $\hat{B}_{44_l}^{idp}$ estimation	***	**	*
Precision of $\hat{B}_{44_q}^{idp}$ estimation	***	*	**
Calculation time (low/*** high)*	*	***	**

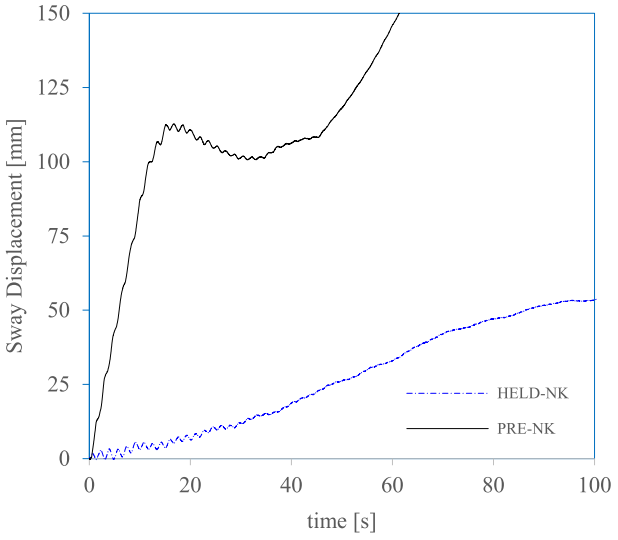


Fig. 11. Sway displacement versus time in the model for Held-Over and Pre-Oscillation tests (see definitions in Fig. 12).

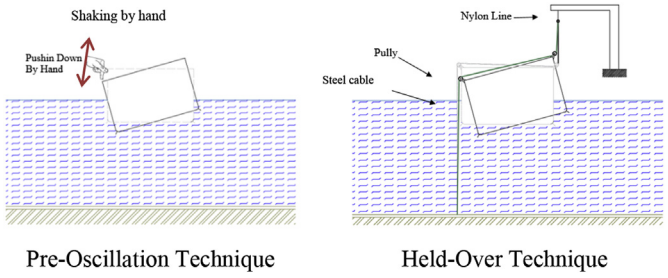


Fig. 12. Schematic of the experimental setup.

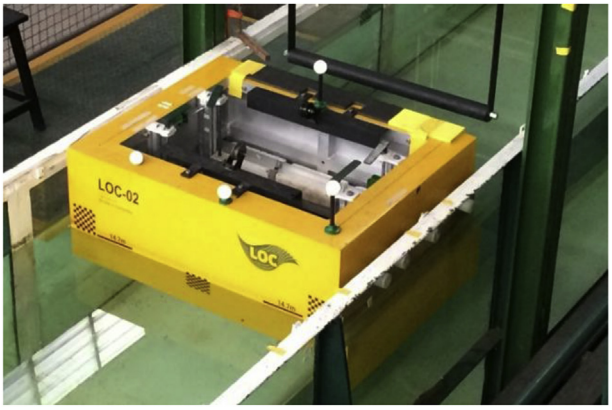


Fig. 13. Model and setup used for the tests.

Table 5
Combination of test conditions.

Test Type	Draft	Natural period (Roll)	Bilge keel	Initial Angle
PRE – (Pre-oscillation) HELD – (Held-Over)	14.7 m (medium)	15 s.	NH – (Naked Hull) BK – (Bilge Keel; L = 1.80 m)	5° 10°. 15°.

6. Consequence of applying the third order equation

6.1. Numerical study

This study now proceeds with an application of the third order equation, from which it is possible to extract several consequences that have been previously overlooked. Calculations of hydrostatic coefficients, added mass and potential damping coefficients were performed using a radiation/diffraction program – WAMIT®. A model with 3703 panels has been used for the representation of the FPSO's hull - see Fig. 6. A common practice is to use extra damping values to allow for the viscous effects when using potential theory. Table 1 presents particular specifications of the full-scale FPSO for a reference draft of 14.7 m.

To verify and validate the system identification approach, time series were generated based on the numerical model's coefficients for different initial conditions (θ_0) and nonlinearity ratios $\frac{\hat{B}_{44_q}^{idp}}{B_{44_l}^{idp}}$ provided by the general solution of the equations using the IDP as a pole. The initial conditions and nonlinearity ratios used are listed in Table 2. The combination of initial conditions and nonlinearity ratios produced a total of 48 conditions for the numerical tests:

The results have shown excellent agreement between the coefficients obtained from system identification and those of numerical models; the differences between them are presented in the next section.

6.1.1. Influence of B_{24} and B_{42} on nonlinear damping estimation by System Identification

6.1.1.1. Quadratic Term. Table 3 shows two different 3rd order SI applications used to study the influence of coupling terms on roll damping. The first case is a 3rd order SI, including $\hat{B}_{24}^{idp}\hat{B}_{42}^{idp}$; the other assumes $\hat{B}_{24}^{idp}\hat{B}_{42}^{idp}$ to be zero.

Fig. 7 shows the consequences of neglecting both $\hat{B}_{24}^{idp}$ and $\hat{B}_{42}^{idp}$ in the calculation of the quadratic term $\left(\hat{B}_{44_q}^{idp}\right)$ for different initial angles. The horizontal axes in the figure are the known values defined in the numerical model, while the vertical axes are the values obtained from the SI.

This sensitivity study reveals a gradual decrease in the precision of the estimation of the quadratic part of roll damping $\left(\hat{B}_{44_q}^{idp}\right)$ when coupling terms are ignored.

6.1.1.2. Linear Term. The numerical errors of calculating the linear damping part ($\hat{B}_{44_l}^{idp}$) for the same cases above ($\hat{B}_{24}^{idp}$ and $\hat{B}_{42}^{idp}$ taken into consideration or assumed to be 0) are compared in Fig. 8.

Generally, 3rd order SI results show good agreement with exact numerical solutions when predicting the linear term of roll damping. For small nonlinearity ratios $\left(\frac{\hat{B}_{44_q}^{idp}}{B_{44_l}^{idp}}\right)$, the value obtained for $B_{44_l}^{idp}$ when including coupling terms in the SI is almost identical to the exact numerical solution; for larger ratios, the error may be up to 1%; however, if coupling terms are ignored in the LSI ($\hat{B}_{24}^{idp}\hat{B}_{42}^{idp} = 0$) the error may increase by up to sevenfold (Fig. 9 (a)).

Finally, Fig. 9 shows the errors of the SI approach for estimating the

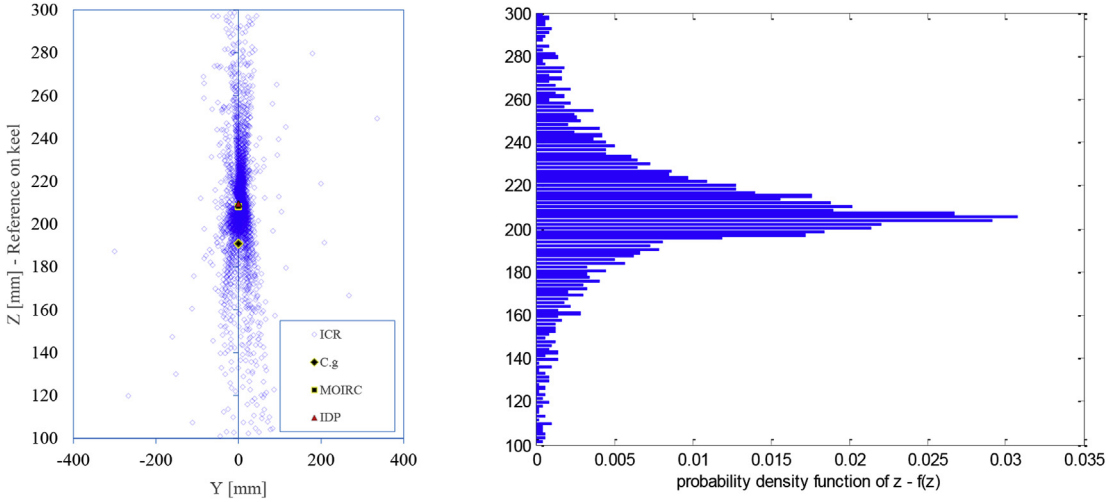


Fig. 14. Measurements of a decay test with the model from Table 6; (a) Comparison of the behavior of the IDP and the MOIRC; (b) Histogram of the vertical position (base point on keel) of the ICRs for model test scale.

Table 6
Particulars of the model used for tests.

Properties	
Displacement	126.69 kg
Length Overall	900 mm
Beam	726 mm
Depth	447 mm
Reference Draft	196 mm
KG	193.1 mm
Natural Period	1.732 s
Bilge keel	0/24 mm

quadratic term of roll damping $\hat{B}_{44_q}^{idp}$, with and without taking $\hat{B}_{24}^{idp}\hat{B}_{42}^{idp}$ into consideration. Increasing nonlinearity leads the 3rd order SI to a minimum error (less than 1%) when the coupling terms are taken into account (Fig. 9(b)); however, the error increases to 30% when the coupling terms are not included in the SI. Note also that Figs. 8 and 9 show no significant change in the error with variation of the initial angle, especially when coupling terms are taken into account.

The comparisons between the numerical study and SI approaches, show a high accuracy of roll damping prediction by 3rd order System Identification when taking coupling terms into account. Therefore, it can be concluded that neglecting the coupling term is a potential source of error for the estimation of roll damping. This is considered an unexpected result given the guidelines given in Lewis (1989).

6.1.2. Comparison of Faltinsen's and Froude's results and 3rd order SI approaches

Previous studies evaluating the nonlinearity of roll damping used Froude's and Faltinsen's approaches for comparisons. This part of the study compare these approaches with the 3rd order SI; considering the results above, coupling terms are included for these simulations. Fig. 10 shows a comparison of the error of roll damping estimation (both linear and quadratic damping terms) between the three approaches.

Froude's approach is very sensitive to both the initial angle and nonlinearity, and leads to larger errors. The errors with Faltinsen's approach are constant for different initial angles, but raise quickly with nonlinearity ratios.

As shown in Fig. 10(a), Froude's approach is unfavorable for predicting linear damping, while Faltinsen's approach is unfavorable for predicting the quadratic roll damping term for systems with high nonlinearity. However, Faltinsen's approach showed reasonable results for predicting linear roll damping with low nonlinearity.

In summary, compared to typical roll damping estimation methods, the 3rd order S.I performed better, showing excellent results for the vast majority of nonlinearity ratios and initial angles. Table 4 provides a brief summary of the comparisons.

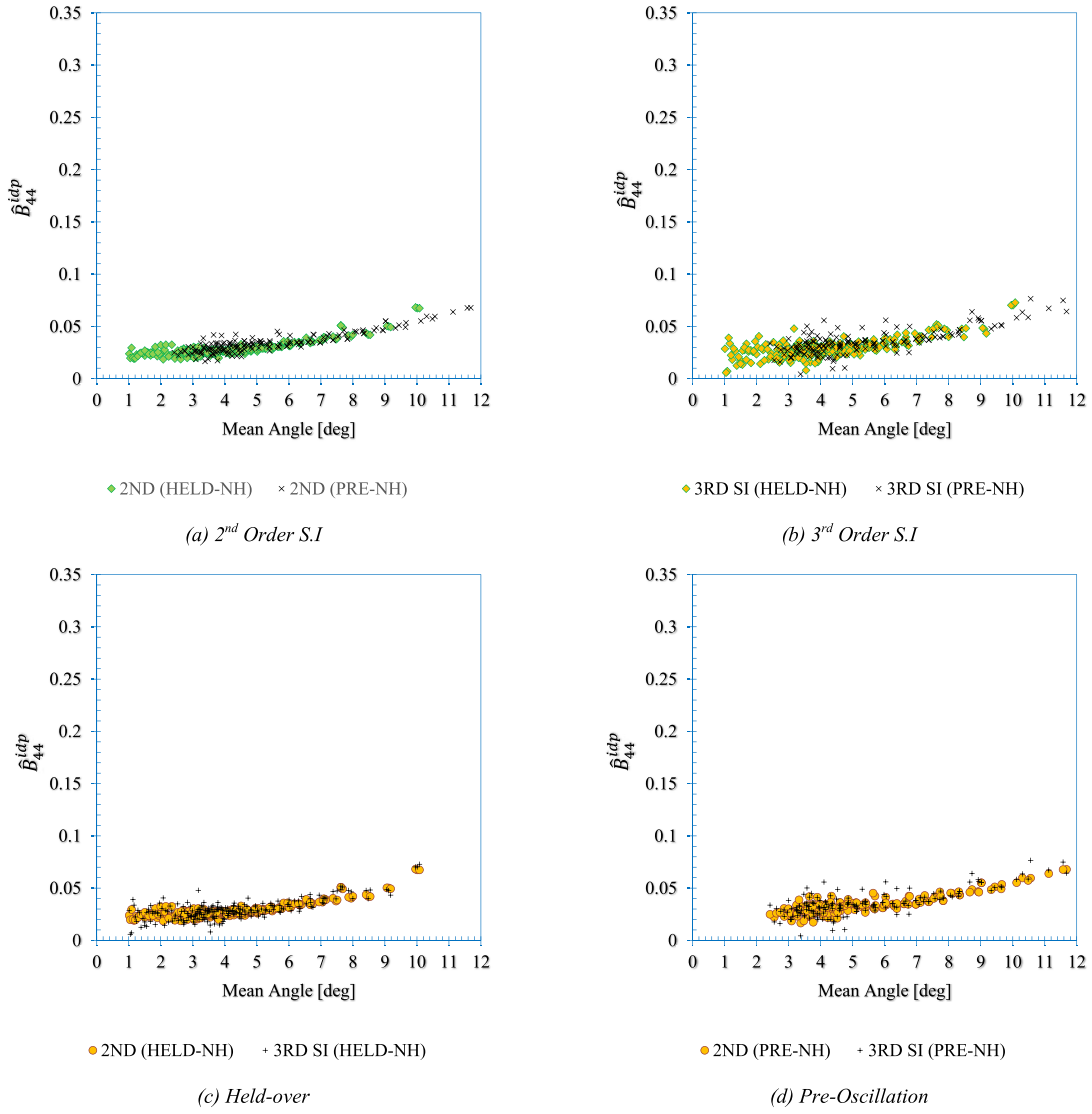
6.2. Experimental study

This study carried out a model test with a section of the parallel middle body of a typical FPSO placed transverse to the Wave Channel width. The width of the Wave Chanel used was 1.00 m, and the width of the model was 0.90 m, leaving a 0.05 m gap at each side. This may be considered a 2D situation, although it is representative for a large parallel middle body FPSO in beam seas (Oliveira and Fernandes, 2014). This has been also checked via potential flow calculations (Asgari and Fernandes, 2017).

Tests were performed under two initial conditions: free roll decay test with pre-oscillation and held-over free roll decay test (see Fig. 12). Due to the coupling terms, sway is unavoidable in any free decay test. Fig. 11 illustrates sway for both free roll decay tests. This is another motivation to take into account the effects of sway for roll damping

Table 7
Different system identification approaches used for experiments.

2nd SI:	$\ddot{\eta}_4^* + \hat{B}_{44_l}^{idp}\dot{\eta}_4^* + B_{44_q}^{idp}\lvert\dot{\eta}_4^*\rvert\dot{\eta}_4^* + \hat{C}_{44}^{idp}\eta_4^* = 0$
3rdSI:	$\ddot{\eta}_4^* + \{\hat{B}_{22} + \hat{B}_{44_l}^{idp}\}\ddot{\eta}_4^* + 2\times\hat{B}_{44_q}^{idp}\lvert\dot{\eta}_4^*\rvert\ddot{\eta}_4^* + \hat{B}_{22}\hat{B}_{44_q}^{idp}\lvert\dot{\eta}_4^*\rvert\dot{\eta}_4^* + \{\hat{B}_{22}\hat{B}_{44}^{idp} - \hat{B}_{24}^{idp}\hat{B}_{42}^{idp}\}\dot{\eta}_4^* + \hat{C}_{44}^{idp}\dot{\eta}_4^* + \hat{B}_{22}\hat{C}_{44}^{idp}\eta_4^* = 0$



(a) 2nd Order S.I *(b) 3rd Order S.I*

(c) Held-over *(d) Pre-Oscillation*

Fig. 15. Comparison of non-dimensional equivalent roll damping of naked hull (NH) model for Held-Over and Pre-Oscillation tests (second and third order approaches).

even under controlled conditions.

The model's hull was made of fiberglass with an internal aluminum structure (see Fig. 13). The test matrix was a combination of pre-oscillation and held-over tests with naked hull and bilge keel models, using three different initial angles in a medium draft (see Table 5). Scaling was based on Froude's similarity with a scale factor of 1:75.

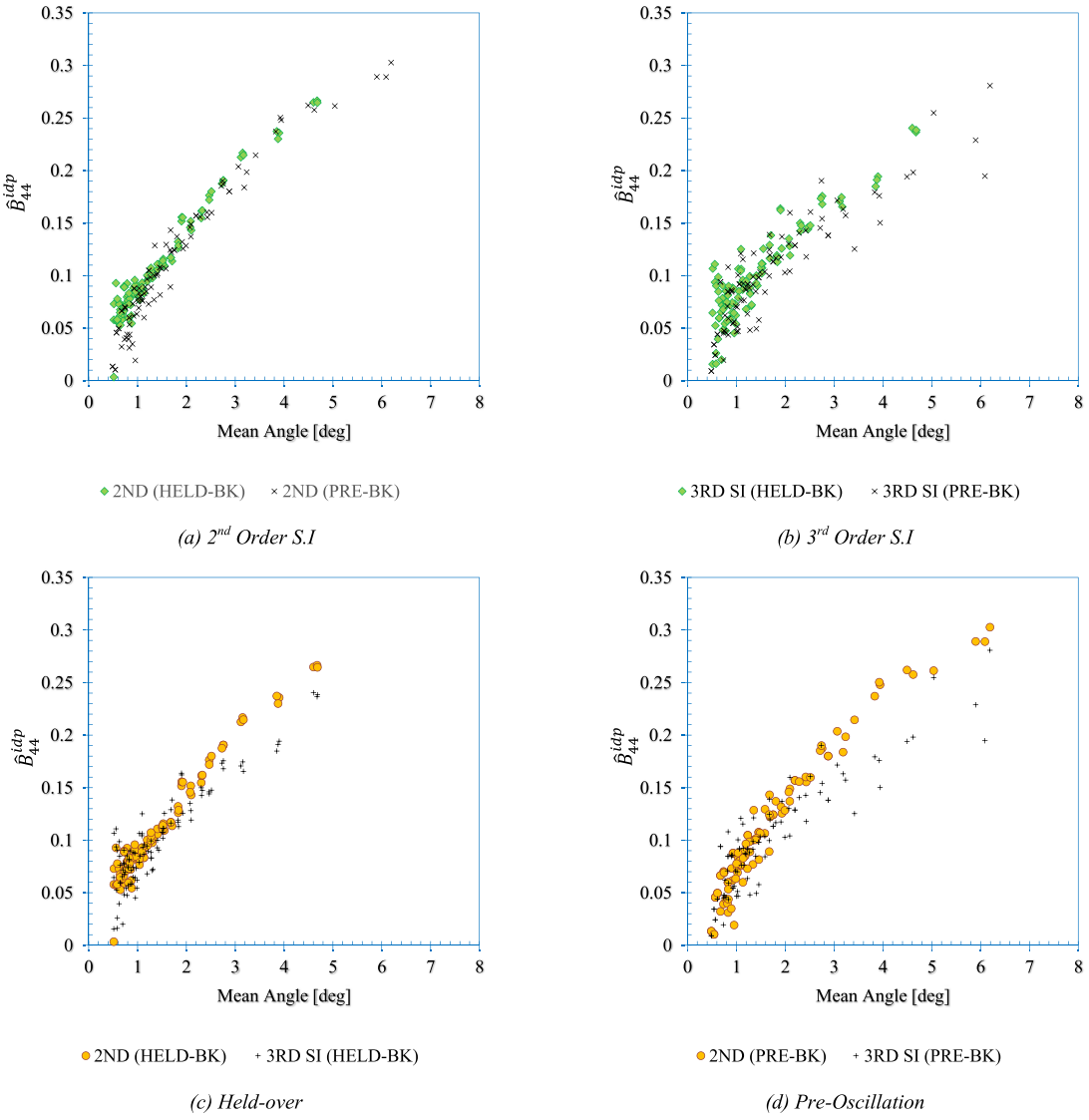
Fig. 14 compares the IDP and the MOIRC (Most Often Instantaneous Roll Center) concepts. Both show almost the same results for free roll decay, but it should be considered that these concepts have different definitions. For different conditions, such as in beam seas, they have completely different behaviors (Fernandes et al., 2016).

As shown in Table 7, two approaches were applied to the experimental results to obtain roll damping coefficients. Non-dimensional roll damping coefficients obtained from 3rd order and 2nd order (used in typical methods) system identification are presented, respectively, in Figs. 15 and 16. Fig. 15(a) and (b) compare non-dimensional roll damping coefficients between the held-over and pre-oscillation tests of the naked hull model, obtained respectively from 2nd and 3rd order system identifications. The comparison shows reasonable agreement of both techniques. These results may provide further support to the hypothesis that a simpler technique of pre-oscillation could be employed instead of applying strict initial conditions for a free roll decay (held-over), but it is important to mention that this is true only for angles when the non-linearity is weak.

Now, Fig. 15(c) and (d) compare separately non-dimensional roll damping coefficients between 2nd and 3rd order system identifications for held-over and pre-oscillation techniques. According to Fig. 15(c) for held-over technique, the results of the 2nd order SI are similar to those of the 3rd order SI for the naked hull model with low nonlinearity. The same conclusion is obtained for the pre-oscillation technique (See Fig. 15(d)).

Observing Fig. 15, one may say that the pre-oscillation technique helps to achieve larger mean angles with the same initial angles since the first cycle can be included when estimating roll damping, however, the pre-oscillation technique shows a larger dispersion than the held-over approach.

As expected, attaching the bilge keel increases the nonlinear damping. Fig. 16(a) and (b) compare non-dimensional roll damping coefficients of the model with a bilge-keel between held-over and pre-oscillation techniques, respectively, obtained from both 2nd and 3rd order system identifications. The comparison shows that the results from both techniques are in good agreement. Fig. 16(c) and (d) show



(a) 2nd Order S.I *(b) 3rd Order S.I*

(c) Held-over *(d) Pre-Oscillation*

Fig. 16. Comparison of non-dimensional equivalent roll damping of the model with a 1.8 m bilge keel (BK) for Held-Over and Pre-Oscillation tests (second and third order approaches).

that the results of held-over have smaller variance than those of the pre-oscillation technique, but it is clear that the 3rd order technique is sensing the non-linearity. There is an overestimation of the damping with a 2nd order technique for large angles (> 2deg).

Based on the results obtained, it may be concluded that more realistic results are to be expected when taking into account the effects of the coupling terms. Fig. 17 shows that a reduction in the quadratic terms of roll damping is observed with the linear-plus-quadratic approach for pre-oscillation tests. Furthermore, using a biquadratic form[1] leads to a better nonlinear damping behavior for larger angles (See Figs. 18 and 19).

With traditional approaches, only the properties of roll damping can be obtained from the free roll decay test. Using the 3rd order SI approach, however, properties of the sway motion and its coupling with roll can also be well estimated (see Equation (3.4)).

This is illustrated in Fig. 20, which shows the estimated non-dimensional form of sway damping ($\hat{B}_{22}$) for tests on models with a naked hull (NH) and with a bilge keel (BK). $\hat{B}_{22}$ showed linear behavior for tests with a naked hull (Fig. 20(a)), but increasing nonlinearity in tests with a bilge keel (Fig. 20(b)).

The product $\hat{B}_{24}^{idp}\hat{B}_{42}^{idp}$ may also be estimated using the 3rd order SI approach. Fig. 21 illustrates the values of the product of damping coupling terms produced by applying this approach to models with a naked hull (NH) and with a bilge keel (BK). The figure shows a clear trend of $\hat{B}_{24}^{idp}\hat{B}_{42}^{idp}$ deviating from constant values within a region near zero, especially for tests on a model with a bilge keel. This clearly illustrates the effect of the bilge keel on the damping coupling terms product which is quite different for the naked hull.

Finally, time series of roll decay modeled with different methods are compared to each other and to experimental results in Fig. 22. Fig. 22(a) shows the time series generated by 3rd order SI approaches with coefficients provided by the cycle-per-cycle analysis; Fig. 22(b) shows the nonlinear roll damping represented in biquadratic form; Fig. 22(c) uses the traditional linear plus quadratic approach; finally, Fig. 22(d) shows the time series for the traditional second-order system identification with a single degree of freedom. Note also that for high

[1] A biquadratic approximation of roll damping divides the mean angle range into two intervals; each interval has different values of linear and quadratic damping, producing two straight lines with different slopes.

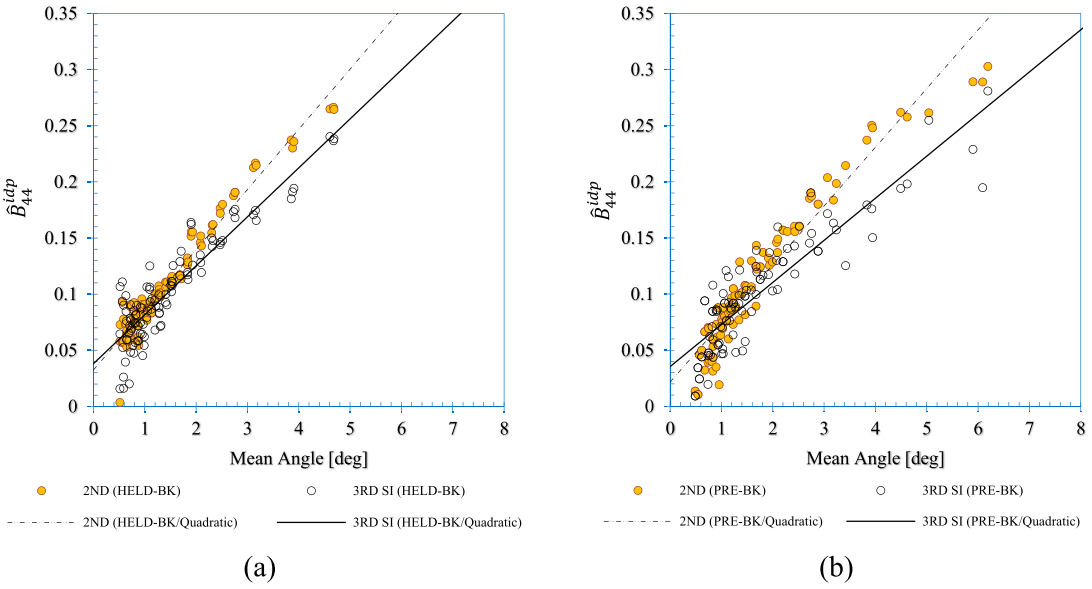


Fig. 17. Linear plus quadratic damping; comparison between second and third order approaches for held-over (a) and pre-oscillation tests (b).

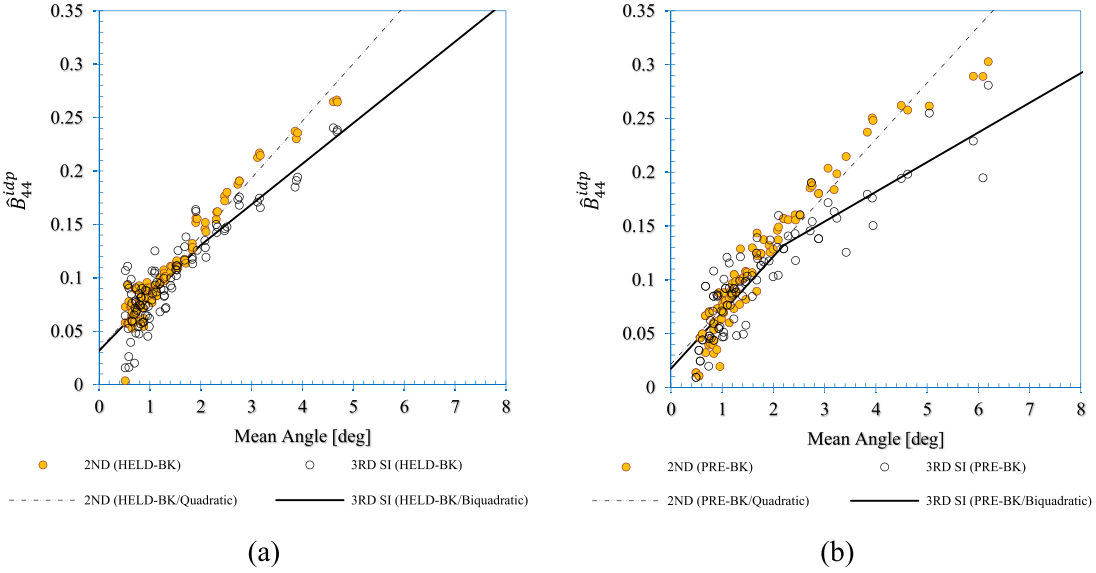


Fig. 18. Comparison between biquadratic damping (3rd order approach) and linear plus quadratic damping (2nd order approach) for held-over (a) and pre-oscillation tests (b).

nonlinearity (bilge keel width equal to 1.8 m), the roll restoring coefficient showed a higher nonlinear deviation.

The influence of coupling terms on the results has already been discussed numerically above; and results have shown in contrast with traditional methods that damping coupling terms are not negligible, especially when roll damping is nonlinear. The two 3rd order SIs shown in Table 3 have also been applied on experimental roll time series to obtain $\hat{B}_{44}^{idp}$; the first included a damping coupling term ($\hat{B}_{24}^{idp}\hat{B}_{42}^{idp} \neq 0$) while in the second the coupling term is neglected ($\hat{B}_{24}^{idp}\hat{B}_{42}^{idp} = 0$). Fig. 23 shows how neglecting coupling terms affects the estimation of $\hat{B}_{44}^{idp}$. Comparing the results of both approaches for a model with a bilge keel (Fig. 23 (b) and (d)), it can be seen that neglecting coupling terms causes overestimation of roll damping for large mean angles, similar to what is seen when applying a 2nd order SI.

The main conclusion here is that the 3rd order SI approach seems to yield better, more robust results when nonlinearity is expected to be higher.

6.2.1. *The role of the coupled damping number - CDN*

In Equation (3.1), the combination of damping terms is always grouped, that is: $\hat{B}_{22}\hat{B}_{44}^{idp} - \hat{B}_{24}^{idp}\hat{B}_{42}^{idp}$. This study therefore proposes that this grouped quantity be defined as a new term, the Coupled Damping Number – CDN, so that

$$CDN \equiv \hat{B}_{22}\hat{B}_{44}^{idp} - \hat{B}_{24}^{idp}\hat{B}_{42}^{idp} \tag{6.2}$$

One should immediately recognize that CDN = 0 (Fernandes, 1983) for the 2D case under analysis under a potential flow hypothesis, which is a result from the application of Green's Theorem.

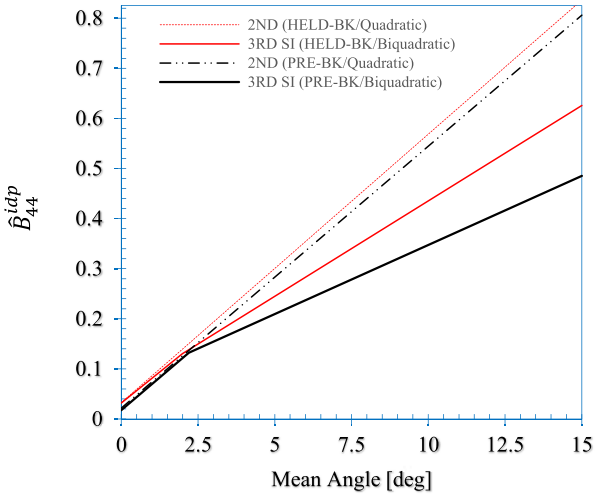


Fig. 19. Direct comparison of the curves for biquadratic damping (3rd order approach - solid lines) and linear plus quadratic damping (2nd order approach - dashed and dotted lines) for held-over (red lines) and pre-oscillation tests (black lines). (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

Hence, Equation (3.1) for this situation can be rewritten as.

$$\ddot{\eta}_4^* + \{\hat{B}_{22} + \hat{B}_{44}^{idp}\}\dot{\eta}_4^* + \{CDN\}\dot{\eta}_4^* + \{\hat{C}_{44}^{idp}\}\dot{\eta}_4^* + \{\hat{B}_{22}\hat{C}_{44}^{idp}\}\eta_4^* = 0 \qquad (6.3)$$

To study the CDN's influence on the equation, two 3rd order SIs were applied on experimental results; the first included the CDN, while the second one neglected it (see Table 8). The results are shown in Fig. 24, below.

Fig. 24 shows that neglecting CDN does not have a large effect on roll damping estimation. This seems to be a remarkable indication that CDN is negligible even when viscous effects are present. It must be stressed that the possibility of neglecting the CDN does not mean that the effects of the coupling damping terms are negligible, but rather that the product of the main diagonal terms minus the product of the coupling terms could indeed be negligible (CDN ≅ 0). This is probably the reason why the wrong assumption that the cross damping term may be neglected has survived for so long!

7. Conclusions

The main aim of the present investigation was to present a robust approach to the assessment of roll damping using an innovative 3rd order equation for roll damping analysis constructed by considering a proper understating of the pole for rotational equations and taking into account the influence of unavoidable coupled dampings. This work points out conditions under which the conventional 2nd Order equation considering only the roll degree of freedom may be used: based on Equation (2.5), sway motion should be very small and, maybe most importantly, the pole should be the IDP (inertially decoupled point) for the 2nd Order equation to work as a good approximation. This study has applied both numerical and experimental techniques. As a first step, it showed the importance of placing the pole at the IDP, which allows a third order equation to be used for roll (Equation (2.7)). The new approach using this third order equation has been validated and verified numerically. The 3rd order SI showed excellent results for the vast majority of nonlinearity ratios and initial angles in the numerical model, and is able to estimate not only roll properties, but also sway and sway/roll coupling properties.

The errors of this 3rd order SI compare favorably with those of the Froude and Faltinsen approaches. The results have shown that the 3rd order SI produces more efficient predictions of the linear-plus-quadratic damping coefficient when compared to Froude and Faltinsen approaches, especially when nonlinearity is high.

This study has shown that including damping coupling terms in 3rd order SIs could improve the estimation of roll damping. Several free roll decay tests using the Held-Over and Pre-Oscillation techniques were carried out in the wave channel facility of the Waves and Currents Laboratory (Laboratório de Ondas e Correntes - LOC) of the Federal University of Rio de Janeiro. The results obtained have shown good agreement of the new approach for different decay techniques. For experiments with a naked hull model, the 2nd order SI has shown similar results to the 3rd order SI; however, in the presence of a bilge keel, the 2nd order SI produces higher roll damping values, while those estimated with the 3rd order SI are more realistic and less unsafe. It is worth mentioning that this approach has produced even better results

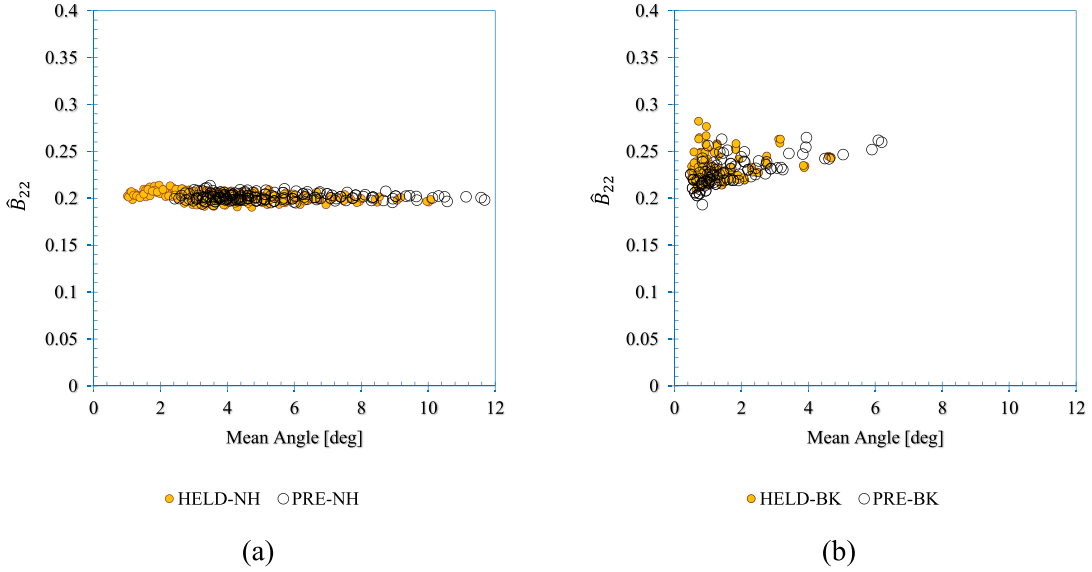


Fig. 20. $\hat{B}_{22}$ estimated in tests with (a) a Naked Hull and (b) a Bilge Keel.

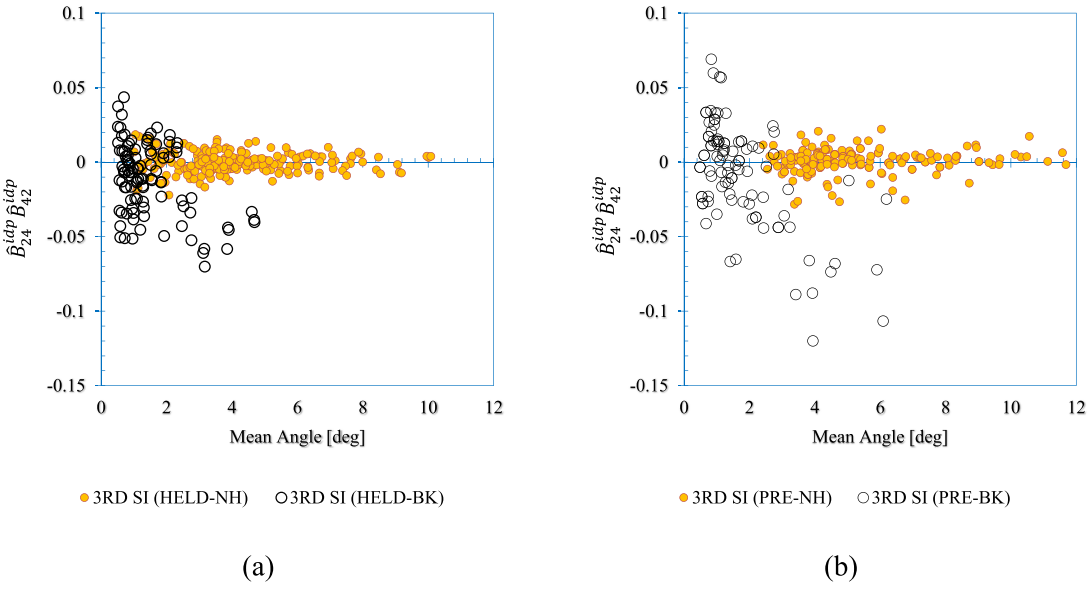


Fig. 21. Non-dimensional form of the product of damping coupling terms for held-over (a) and pre-oscillation (b) tests.

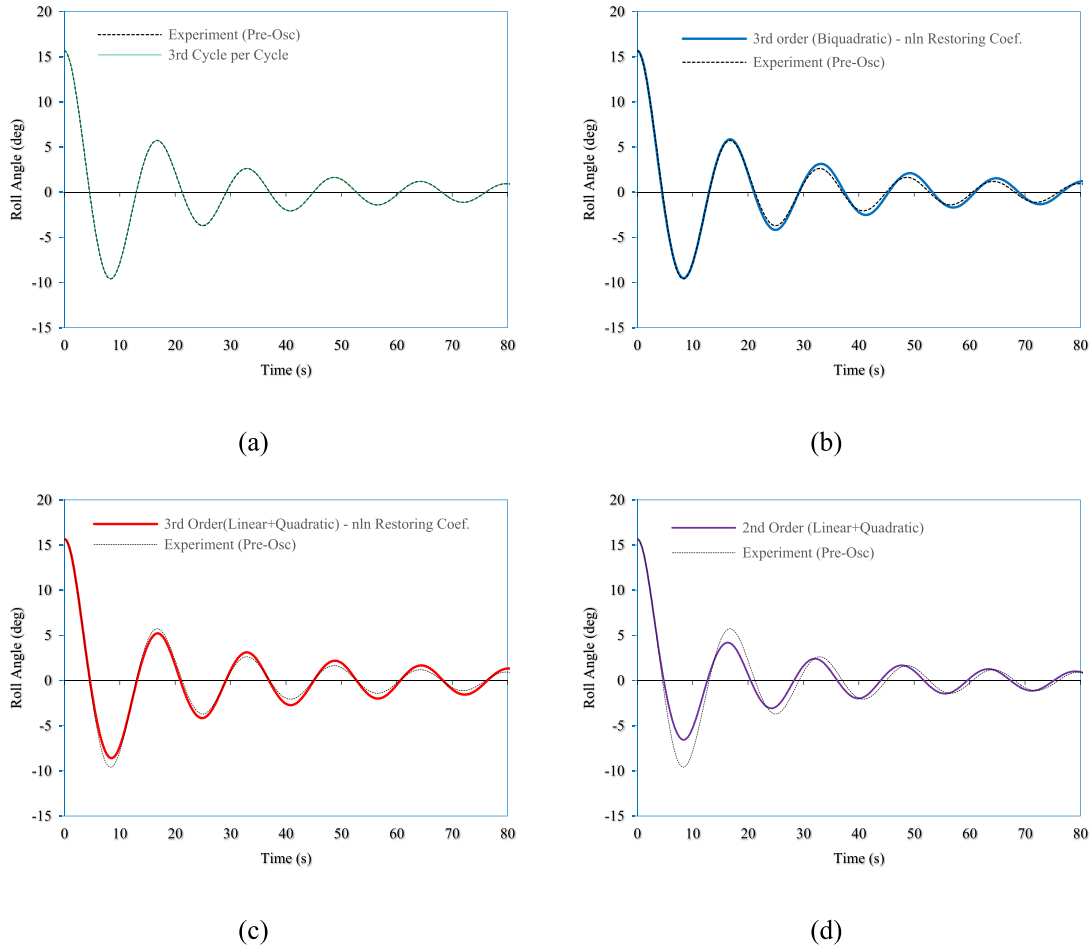


Fig. 22. Roll time series for pre-oscillation produced with different damping approaches.

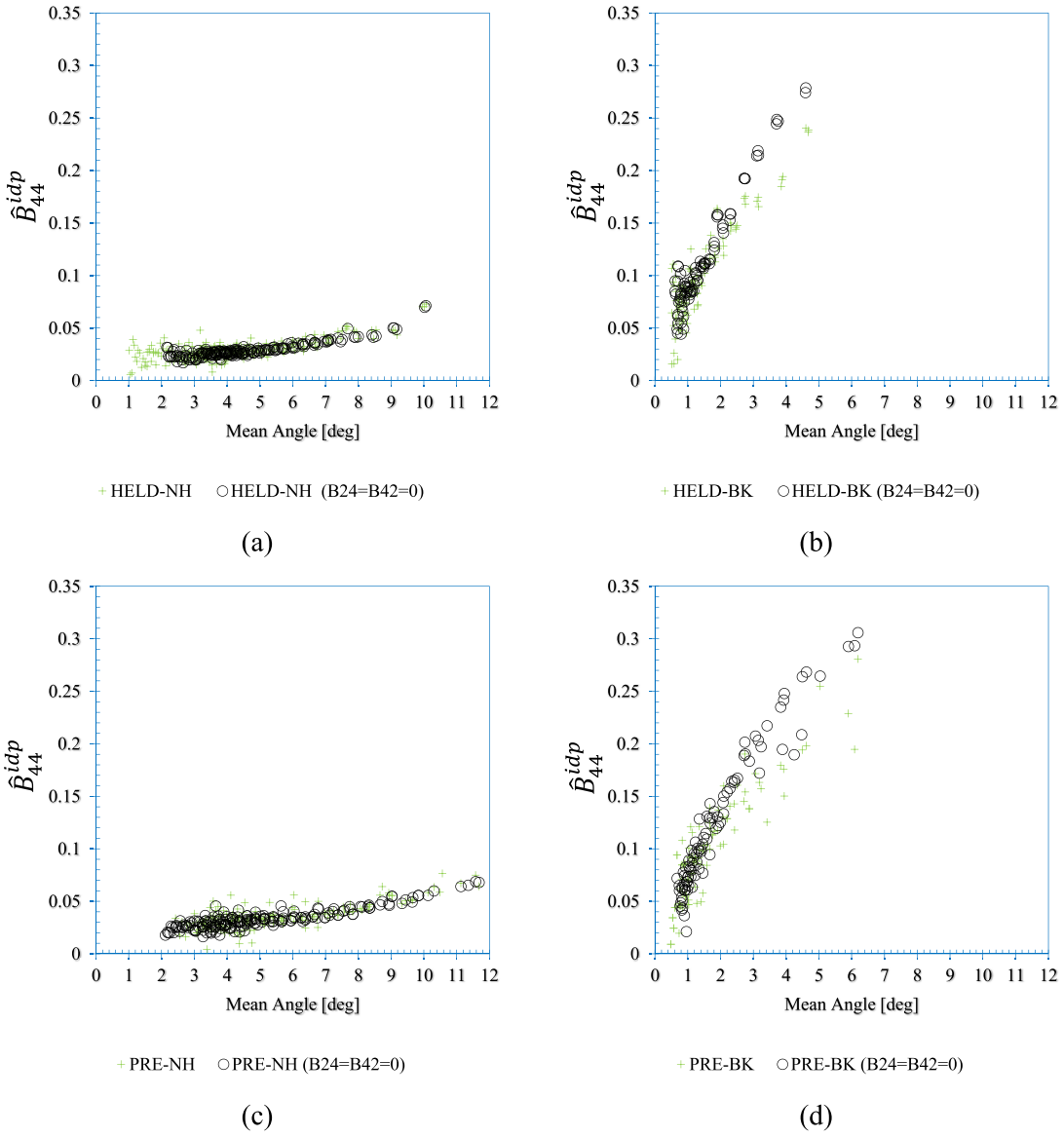


Fig. 23. Equivalent roll damping for ($\hat{B}_{24}^{idp}\hat{B}_{42}^{idp} = 0$) and. ($\hat{B}_{24}^{idp}\hat{B}_{42}^{idp} \neq 0$)

Table 8
SIs including and ignoring the CDN.

CDN ≠ 0	$\ddot{\eta}_4^* + \{\hat{B}_{22} + \hat{B}_{44}^{idp}\}\dot{\eta}_4^* + \{CDN\}\eta_4^* + \{\hat{C}_{44}^{idp}\}\dot{\eta}_4^* + \{\hat{B}_{22}\hat{C}_{44}^{idp}\}\eta_4^* = 0$
CDN = 0	$\ddot{\eta}_4^* + \{\hat{B}_{22} + \hat{B}_{44}^{idp}\}\dot{\eta}_4^* + \{\hat{C}_{44}^{idp}\}\dot{\eta}_4^* + \{\hat{B}_{22}\hat{C}_{44}^{idp}\}\eta_4^* = 0$

for Held-Over tests (see Fig. 23).

Finally, a new CDN (coupled damping number) term including all damping coefficients (Equation (6.2)) is presented here. One of the most significant findings from this study is that the CDN may be negligible even when viscous effects are present (non-potential). As mentioned above, it should be noted that neglecting the CDN does not mean that the effects of the coupling damping term are negligible, but rather that the product of the main diagonal terms minus the product of the coupling terms could indeed be negligible (CDN ≅ 0). This is probably the reason why the wrong assumption that the cross damping term may be neglected has survived for so long! Proving that for the general case is a task for future work.

Acknowledgements

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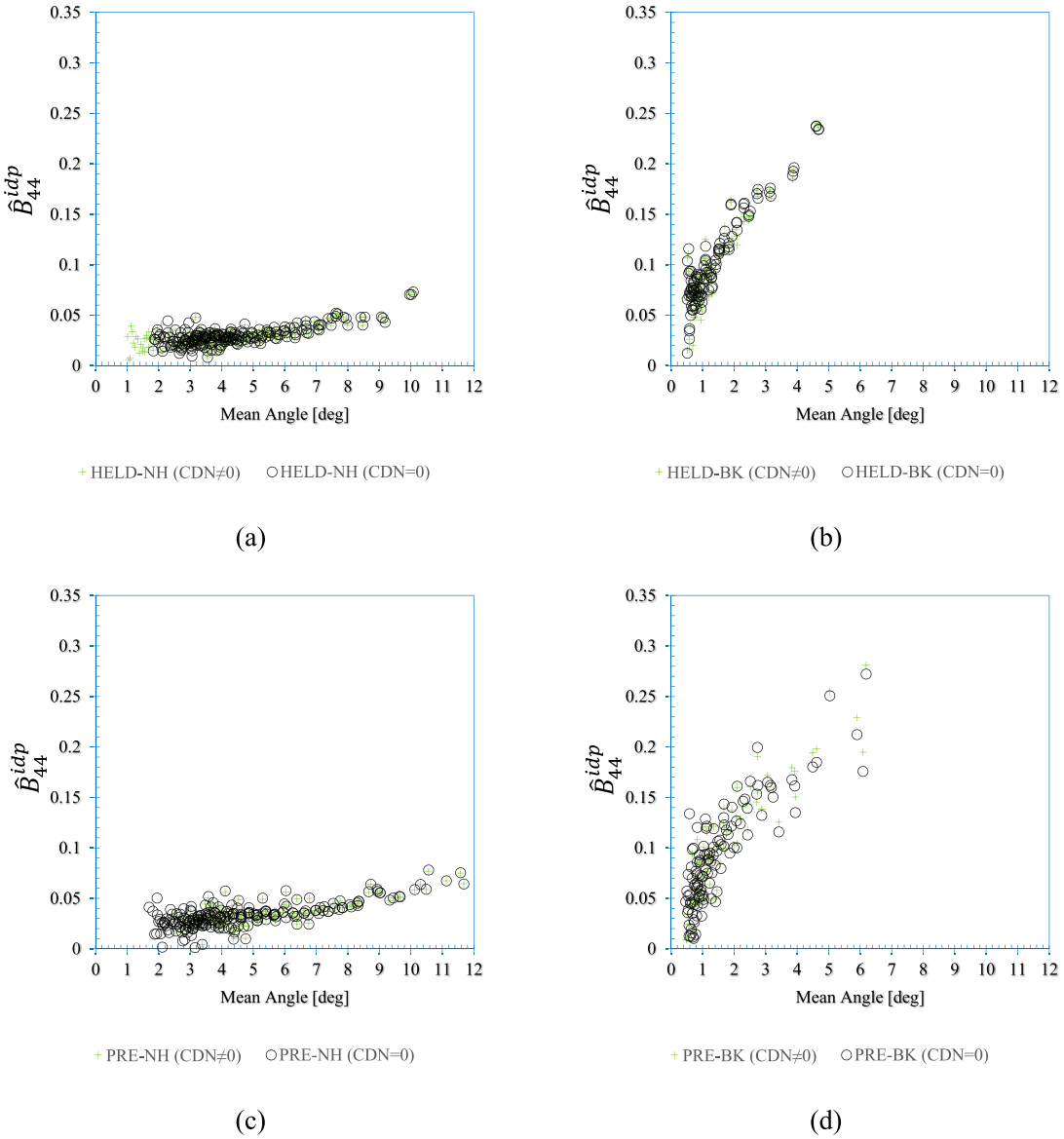


Fig. 24. Comparison of estimated equivalent roll damping for CDN = 0 and CDN ≠ 0 for held-over (a, b) and pre-oscillation tests (c, d) with a naked heel (a, c) and with a bilge keel (b, d).

Appendix A

Motion equations at the IDP can be reduced to the following dimensionless form:

$$(M + A_{22})\ddot{\eta}_2 + B_{22}\dot{\eta}_2 + B_{24}^{idp}\dot{\eta}_4 = 0 \tag{A.1}$$

$$(I_4^{idp} + A_{44}^{idp})\ddot{\eta}_4 + B_{42}^{idp}\dot{\eta}_2 + B_{44}^{idp}\dot{\eta}_4 + C_{44}^{idp}\eta_4 = 0 \tag{A.2}$$

from (A.2):

$$\dot{\eta}_2 = \frac{-1}{B_{42}^{idp}}\{(I_4^{idp} + A_{44}^{idp})\ddot{\eta}_4 + B_{44}^{idp}\dot{\eta}_4 + C_{44}^{idp}\eta_4\} \tag{A.3}$$

Calculating the derivative of (A.3):

$$\ddot{\eta}_2 = \frac{-1}{B_{42}^{idp}}\{(I_4^{idp} + A_{44}^{idp})\dddot{\eta}_4 + B_{44}^{idp}\ddot{\eta}_4 + C_{44}^{idp}\dot{\eta}_4\} \tag{A.4}$$

Substituting (A.3) in (A.1):

$$\ddot{\eta}_2 = \frac{B_{22}}{(M + A_{22})}\{(I_4^{idp} + A_{44}^{idp})\ddot{\eta}_4 + (B_{44}^{idp} - B_{24}^{idp}B_{42}^{idp})\dot{\eta}_4 + C_{44}^{idp}\eta_4\} = 0 \tag{A.5}$$

Substituting Equation (A.5) in Equation (A.4), the 3rd order ordinary differential homogenous equation for roll motion with two degrees of

freedom can be expressed as:

$$\{I_4^{idp} + A_{44}^{idp}\}\ddot{\eta}_4 + \left\{\frac{B_{22}(I_4^{idp} + A_{44}^{idp})}{(M + A_{22})} + B_{44}^{idp}\right\}\ddot{\eta}_4 + \left\{\frac{B_{22}B_{44}^{idp} - B_{24}^{idp}B_{42}^{idp}}{(M + A_{22})} + C_{44}^{idp}\right\}\dot{\eta}_4 + \left\{\frac{B_{22}C_{44}^{idp}}{(M + A_{22})}\right\}\eta_4 = 0 \tag{A.6}$$

The 3rd order ordinary differential non-homogenous equation for sway motion forced by roll can be obtained following the same procedure described above.

$$\{I_4^{idp} + A_{44}^{idp}\}\ddot{\eta}_2 + \left\{\frac{B_{22}(I_4^{idp} + A_{44}^{idp})}{(M + A_{22})} + B_{44}^{idp}\right\}\ddot{\eta}_2 + \left\{\frac{B_{22}B_{44}^{idp} - B_{24}^{idp}B_{42}^{idp}}{(M + A_{22})}\right\}\dot{\eta}_2 = \left\{\frac{B_{22}C_{44}^{idp}}{(M + A_{22})}\right\}\eta_4 \tag{A.7}$$

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